

DS 203 – Big Data Essentials
Advanced SQL Lab 2 – Indexes
Marks: 100

Student ID:

Student Name:

Answer the following questions:

Q1. Create an index on the 'email' column in 'customer'.

Query:

Output Screenshot:

Q2. Drop the index created in Q1 and observe how performance changes. Look for impact on execution time and planning time.

Query 1:

First, before dropping the index write the query to extract information from customer table based on email use EXPLAIN ANALYZE command to show execution and planning times:

Output Screenshot:

Query 2:

Second, drop the index:

Output Screenshot:**Query 3:**

Third, after dropping the index write the query to extract information from customer table based on email again and use EXPLAIN ANALYZE command to show execution and planning times after dropping index:

Output Screenshot:

Q3. Create a compound index on 'first_name, last_name' in the actor table

Query:

Output Screenshot:

Q4. Use pg_indexes to list all indexes on the 'actor' table.

Query:

Output Screenshot:

Q5. Try indexing a column with few unique values like “active” column in “customers” table and compare the performance - does it help? Explain your answer.

Answer:

Marking Criteria:

For each question from Q1-Q4:

Correct Query = 10 Marks

Correct Output Screenshot = 10 Marks

For Q5: 20 marks.

Total Marks: 20 x 5 (questions) = 100 Marks.

Submission Instructions:

Submit the following:

1. Convert this document into PDF and rename it as “**DS 203 - Lab 2 – StudentName.pdf**” then submit in the appropriate submission on Google Classroom.
2. Also, submit the .sql script containing queries for Q1-Q4 in single file. You can name the script file as “**DS 203 - Lab 2 – StudentID.sql**”.